

Ex No: 9

LOW PASS FILTER AND HIGH PASS FILTER.

10/9/19

AIM:

To design and test the frequency response of a second order. low pass filter and high pass filter.

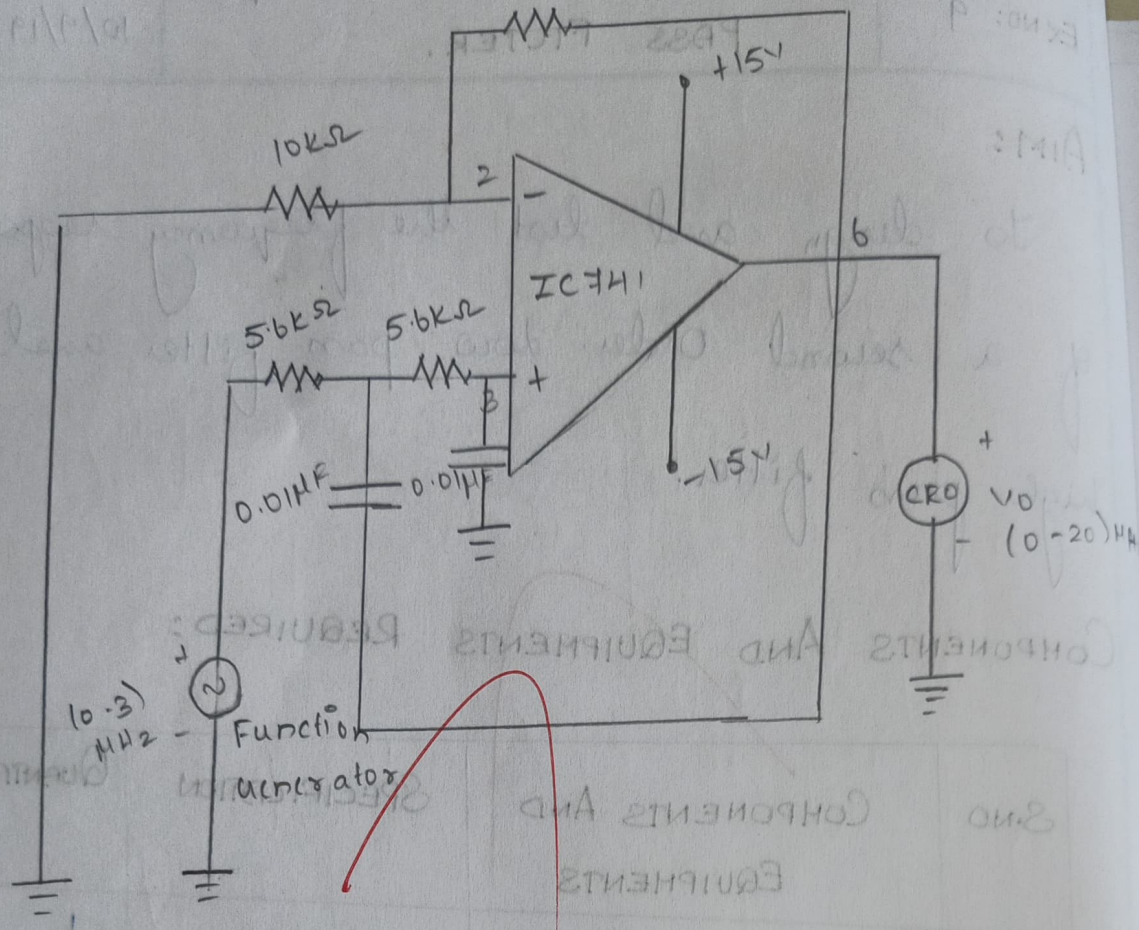
COMPONENTS AND EQUIPMENTS REQUIRED:

S.NO	COMPONENTS AND EQUIPMENTS	SPECIFICATION	QUANTITY
1.	OP - Amp	IC 741	1
2.	Resistors	10k Ω , 5.6k Ω	2
3.	Capacitor	0.01 μ F	1
4.	RPS	(0-30)V	1
5.	CRO	(0-20) MHz	1
6.	Function generator	(0-2) MHz	1
7.	Connecting wires	-	As required.

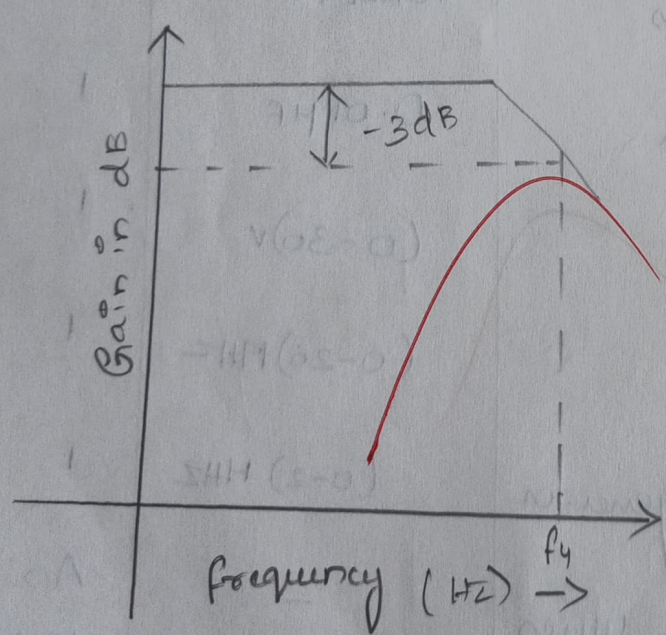


Low Pass Filter

CIRCUIT DIAGRAM



Model Graph:-



f_H - cut off frequency (HPF)

f_L - cut off frequency (LPF).

THEORY:

An active filter is a type of analog electronic filter that uses active components such as amplifiers. Amplifiers included in a filter design can be used to improve the performances and predictability of a filter, while avoiding the need for affecting the characteristics of the filter. The shape of the response, the tuned frequency can often be set with inexpensive variable resistors. In some active filter circuits.

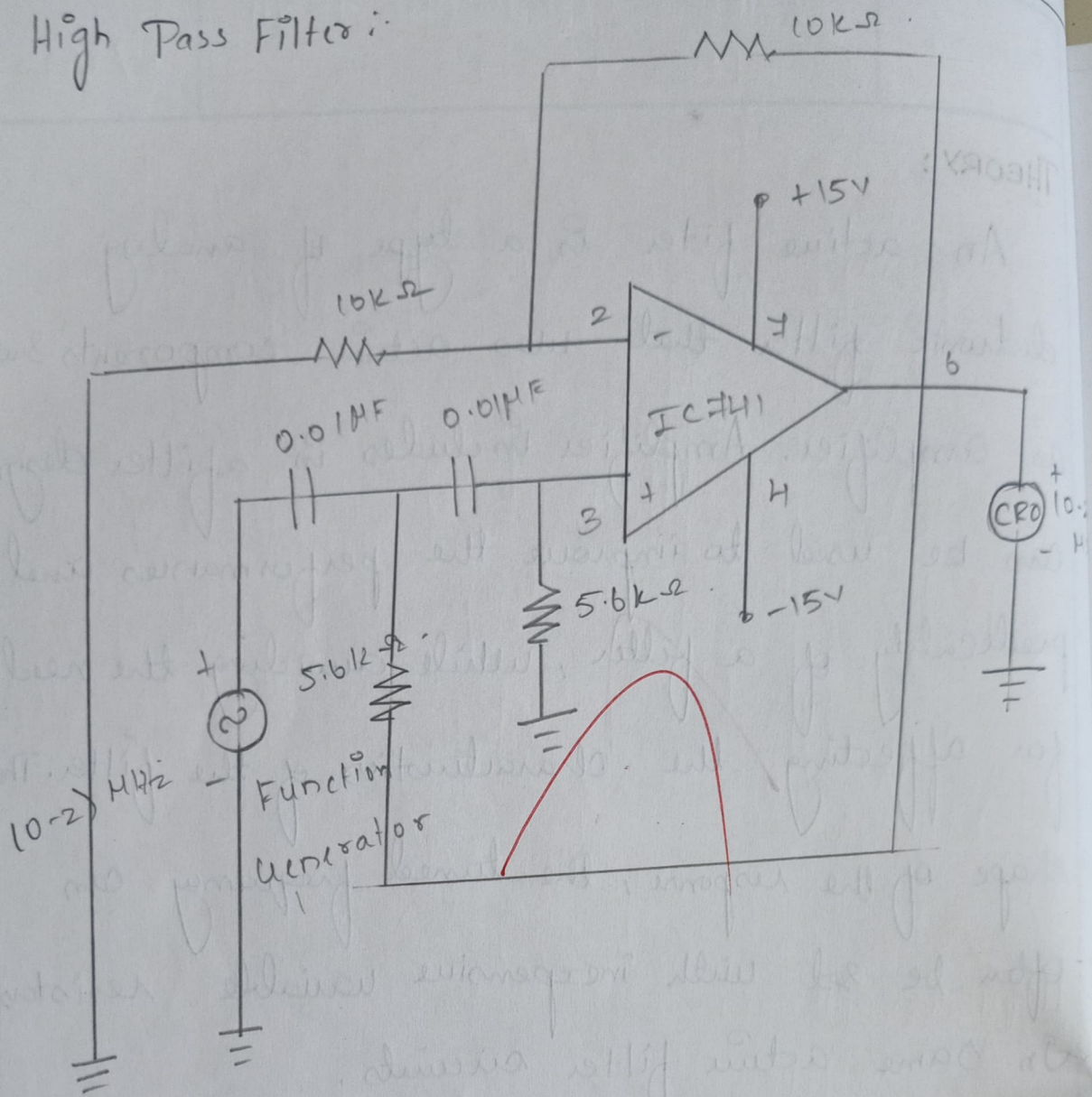
Second order active filter whether low pass or high pass, are important in electronics because we can use them to design much.

LOW PASS FILTER:

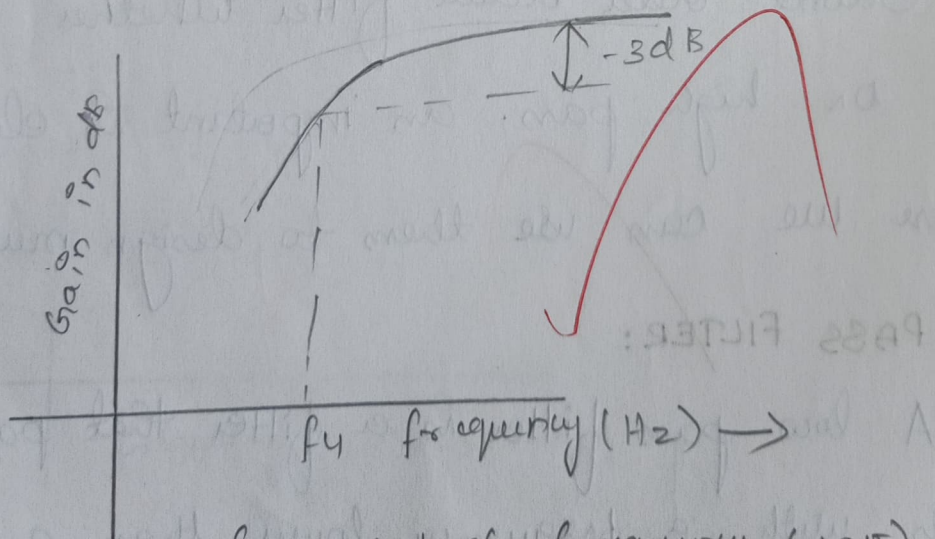
A low pass filter is a filter that passes signals with a frequency lower than a certain cut-off frequency and attenuates signals with



High Pass Filter:



Model Graph:-



f_c - cut off frequency (HPPF)

f_c - cut off frequency (LPPF).

frequencies higher than the cut-off frequency.

This second order low pass filter circuit has two RC network, $R_1 - C_1$ and $R_2 - C_2$ which gives the filter its frequency response proportion. The filter design is based around a non-inverting OP-Amp configuration so that filter again. A will always be greater than 1, also the OP-amp has a high input.

HIGH PASS FILTER:

A filter of high pass filter is an electronic filters that passes signals with a frequency higher than a certain cut-off frequency and attenuates signal with frequency lower than the cut-off frequency. The amount of attenuation for each frequency depends on the filter, a first order high pass active filter can be converted into



LOW PASS Filter

TABULATION :-

Frequency (Hz)	OUTPUT voltage (V _o)	Gain = $\left(\frac{V_o}{V_{in}}\right)$	Gain in dB = $20 \log(V_o/V_{in})$
200 Hz	0	0	0
500 Hz	0	0	0
900 Hz	0	0	0
800 Hz	3.4 x 20	5.66	15.06
1 KHz	8.6 x 20 V	6.00	15.58
1.2 KHz	3.8 x 20 V	6.33	16.03
3 KHz	4 x 20 V	6.66	16.47
5 KHz	4 x 20 V	6.66	16.47
8 KHz	4 x 20 V	6.66	16.47
10 KHz	4 x 20 V	6.66	16.47

a second order high pass filter first supply by using an association DC network in the input path. The frequency response of the second-order high pass filter is identical to that of the first order filter at 40dB/decade.

PROCEDURE:

Connections are given as per the circuit diagram.

Input signal is given using signal generator.

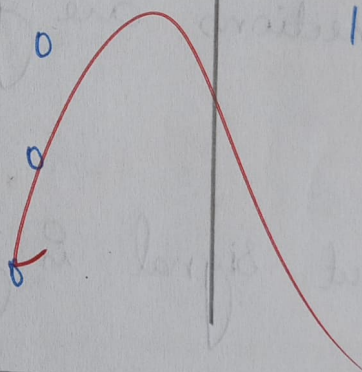
Varying the frequency of the input signal from 100 Hz to 1 MHz and obtain the corresponding amplitude variation of the output signal.

Tabulate the readings.

Plot the graph with frequency and gain.

HIGH PASS FILTER

Frequency (Hz)	Output voltage (V _o)	Gain = (V_o/V_{in})	gain in dB $= 20 \log (V_o/V_{in})$
200Hz	4.8 x 2V	8	18.06
300Hz	4.8 x 2V	8	18.06
900Hz	4.8 x 2V	8	18.06
1KHz	4.5 x 2V	7.5	17.50
1.5KHz	4 x 2V	6.6	16.39
2KHz	3.9 x 2V	6.5	16.25
25KHz	0	0	0
3KHz	0	0	0
3.5KHz	0	0	0



gain (f dB) →

105

dB ↓ -3dB

~~Roll off~~

low pass filter

Scale
x axis 1 decade = 10¹ Hz

n = 1, 2, 3, ...

y axis 1 cm = 5 dB

Cut off frequency = 2 kHz

Bandwidth = $f_H - f_L$

15
10
5
0

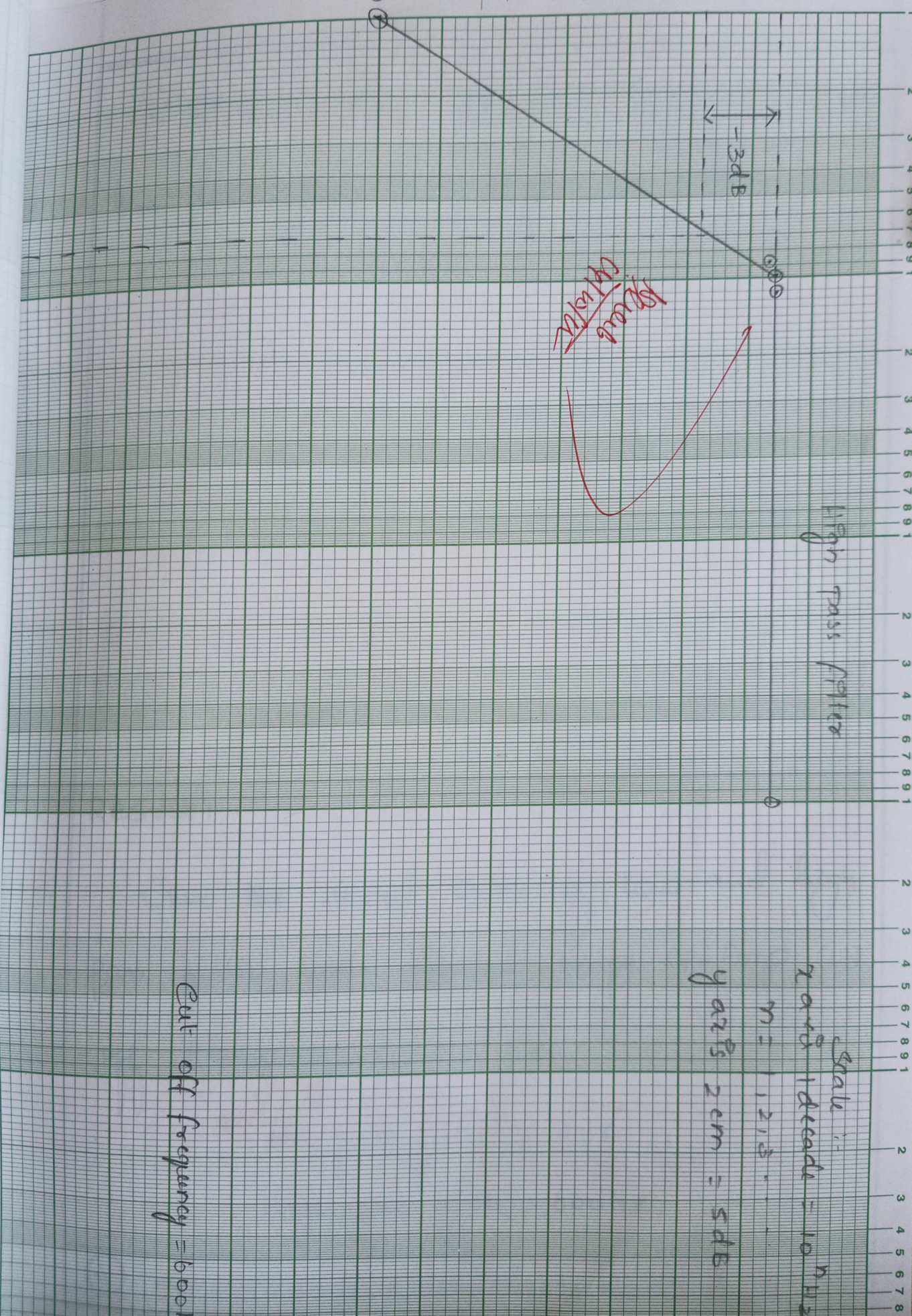
-3dB

~~High pass filter~~

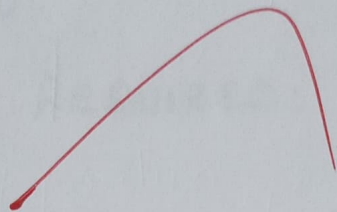
High pass filter

Scale :-
x axis 1 decade = 10 Hz
y axis 2 cm = 5 dB

Cut off frequency = 600 Hz



Determine the cut-off frequency for low pass filter and high pass filter.



RESULT:

Thus the second order of low and high pass filter was divided using op-Amp cut-off frequency.

~~200Hz~~
low pass filter = 2KHz

High pass filter = 600Hz.

